Lesson 49: SARSA – On-Policy TD Control

AI/ML Mathematics

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1 Motivation and Overview

Lessons 47 and 48 developed temporal-difference methods for *prediction*: estimating the value function V^π of a fixed policy π. But the ultimate goal of reinforcement learning is *control* — finding a policy that maximizes cumulative reward. Consider a robot navigating a factory floor, where each wrong turn costs time and battery. The robot cannot afford to evaluate every candidate policy to completion; it must improve its behavior *while interacting with the environment*. This requires simultaneously evaluating the current policy and improving it — a challenge known as the *exploration–exploitation dilemma*.

The transition from prediction to control introduces two fundamental difficulties. First, the agent must learn *action-value* functions $Q^\pi(s, a)$ rather than state-value functions $V^\pi(s)$, because computing the greedy policy from V^π requires a model of the environment, while computing it from Q^π does not. Second, the agent must maintain exploration to ensure all state–action pairs are sampled, while simultaneously converging to a greedy policy. SARSA resolves both difficulties by applying the TD(0) update to action-value functions under an ε-greedy policy that becomes greedier over time. The tools developed in this lesson will let us formalize the SARSA algorithm, state the GLIE convergence conditions, and prove that SARSA converges to the optimal action-value function Q^*.

Where SARSA and On-Policy Control Appear

- **SARSA(λ)** extends SARSA with eligibility traces (Lesson 48), propagating TD errors backward through recently visited state–action pairs for faster credit assignment.
- **Expected SARSA** (Lesson 50) replaces the sampled next action with an expectation, reducing variance while maintaining the on-policy structure.
- **Actor–critic methods** generalize the on-policy evaluation–improvement loop of SARSA to the function approximation setting, using a critic (value estimator) and an actor (policy).
- **Safe reinforcement learning**: SARSA's on-policy nature means it accounts for the cost of exploration, often yielding safer behavior than off-policy alternatives like Q-learning.

Prerequisites. TD(0) update rule and TD error δ_t (Lesson 47, Definitions 4.1–4.2). Convergence of TD(0) via stochastic approximation: the Bellman operator $\mathcal{T}^\pi$ as a γ-contraction, martingale noise, and the Robbins–Monro theorem (Lesson 47, Theorem 4.7). Bellman expectation and optimality equations for action-value functions Q^π and Q^* (Lessons 45–46). Step-size conditions $\sum_n \alpha_n = \infty$, $\sum_n \alpha_n^2 < \infty$ (Lesson 31).

Learning objectives.

- *Define* the action-value Bellman operator and verify that it is a γ-contraction in $\|\cdot\|_\infty$.
- *State* the SARSA update rule and explain the acronym $(S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1})$.
- *Define* ε-greedy policies and the GLIE (Greedy in the Limit with Infinite Exploration) condition.
- *Prove* the convergence of SARSA to Q^* under GLIE by decomposing the update into a Q-learning component plus a vanishing exploration perturbation.
- *Compute* SARSA updates on small MDPs and compare the learned values with those of a purely greedy agent.

- *Explain* why the on-policy nature of SARSA leads to different behavior than off-policy Q-learning in stochastic environments.

2 References

- Sutton and Barto [2018, Chapter 6, Sections 6.4–6.5] — SARSA, on-policy TD control, and the cliff-walking example comparing SARSA and Q-learning.
- Bertsekas and Tsitsiklis [1996, Chapter 4] — Convergence of TD control methods via stochastic approximation.
- Singh et al. [2000] — The definitive convergence proof for single-step on-policy RL algorithms, including SARSA under GLIE.
- Szepesvári [2010, Section 4.2] — Compact treatment of TD control convergence.
- Rummery and Niranjan [1994] — The original paper introducing SARSA (under the name "modified connectionist Q-learning").
- Tsitsiklis [1994] — Asynchronous stochastic approximation framework underlying the convergence analysis.

3 From Prediction to Control

The TD prediction methods of Lessons 47–48 estimate V^π for a fixed policy π. To find an optimal policy, we employ the *Generalized Policy Iteration* (GPI) framework: alternate between *policy evaluation* (estimating a value function) and *policy improvement* (making the policy greedy with respect to the estimated values).

Definition 3.1 (Generalized Policy Iteration)**.** *Generalized Policy Iteration (GPI)* refers to any algorithm that interleaves partial policy evaluation with policy improvement. The evaluation step updates the value function toward Q^π (or V^π), and the improvement step adjusts the policy to be greedier with respect to the current value estimate.

For model-free control, we must work with action-value functions rather than state-value functions.

Remark 3.2 (Why Action Values?). Given $V^\pi(s)$, selecting the greedy action requires solving $\arg\max_a \sum_{s'} \mathcal{P}(s' \mid s,a)[r(s,a,s') + \gamma V^\pi(s')]$, which requires knowledge of the transition kernel $\mathcal{P}$. In contrast, given $Q^\pi(s,a)$, the greedy action is simply $\arg\max_a Q^\pi(s,a)$ — no model needed. This is why all model-free TD control methods learn Q-values.

3.1 Action-Value Bellman Operators

Definition 3.3 (Action-Value Bellman Expectation Operator)**.** For a policy π, the *action-value Bellman expectation operator* $\mathcal{T}_Q^\pi \colon \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \to \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is defined by

$$(\mathcal{T}_Q^\pi Q)(s,a) = \sum_{s'} \mathcal{P}(s' \mid s,a) \left[r(s,a,s') + \gamma \sum_{a'} \pi(a' \mid s') Q(s',a') \right]. \tag{1}$$

Its unique fixed point is Q^π.

Definition 3.4 (Bellman Optimality Operator on Q-Values)**.** The *Bellman optimality operator* on action values is

$$(\mathcal{T}_Q^* Q)(s,a) = \sum_{s'} \mathcal{P}(s' \mid s,a) \left[r(s,a,s') + \gamma \max_{a'} Q(s',a') \right]. \tag{2}$$

Its unique fixed point is Q^*.

Lemma 3.5 (Contraction of Action-Value Bellman Operators)**.** *Both $\mathcal{T}_Q^\pi$ and $\mathcal{T}_Q^*$ are γ-contractions in $\|\cdot\|_\infty$:*

$$\|\mathcal{T}_Q^\pi Q_1 - \mathcal{T}_Q^\pi Q_2\|_\infty \le \gamma \|Q_1 - Q_2\|_\infty, \qquad \|\mathcal{T}_Q^* Q_1 - \mathcal{T}_Q^* Q_2\|_\infty \le \gamma \|Q_1 - Q_2\|_\infty. \tag{3}$$

Proof. We prove the result for $\mathcal{T}_Q^\pi$; the argument for $\mathcal{T}_Q^*$ is analogous (using $|\max_a f(a) - \max_a g(a)| \le \max_a |f(a) - g(a)|$). For any (s,a):

$$\begin{aligned} |(\mathcal{T}_Q^\pi Q_1)(s,a) - (\mathcal{T}_Q^\pi Q_2)(s,a)| &= \gamma \left| \sum_{s'} \mathcal{P}(s' \mid s,a) \sum_{a'} \pi(a' \mid s') \big[Q_1(s',a') - Q_2(s',a') \big] \right| \\ &\le \gamma \sum_{s'} \mathcal{P}(s' \mid s,a) \sum_{a'} \pi(a' \mid s') |Q_1(s',a') - Q_2(s',a')| \\ &\le \gamma \|Q_1 - Q_2\|_\infty \underbrace{\sum_{s'} \mathcal{P}(s' \mid s,a) \sum_{a'} \pi(a' \mid s')}_{=1}. \end{aligned} \tag{4}$$

Taking the supremum over (s,a) gives the result. □

The contraction properties established in Lemma 3.5 will be the key stability condition in the convergence proofs that follow.

4 Exploration and ε-Greedy Policies

Definition 4.1 (ε-Greedy Policy)**.** Given an action-value function Q, the *ε-greedy policy* assigns

$$\pi_\varepsilon(a \mid s) = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{|\mathcal{A}|} & \text{if } a = \arg\max_{a'} Q(s,a'), \\ \frac{\varepsilon}{|\mathcal{A}|} & \text{otherwise}, \end{cases} \tag{5}$$

where $\varepsilon \in (0,1]$ and ties in $\arg\max$ are broken arbitrarily but consistently.

Proposition 4.2 (ε-Greedy Policy Improvement)**.** *Let $Q = Q^\pi$ for some policy π, and let π_ε be the ε-greedy policy with respect to Q. Then $Q^\pi(s,\pi_\varepsilon) \ge V^\pi(s)$ for all s, where $Q^\pi(s,\pi_\varepsilon) = \sum_a \pi_\varepsilon(a \mid s) Q^\pi(s,a)$.*

Proof. Since π_ε places probability at least $\varepsilon/|\mathcal{A}|$ on every action and the remaining $1-\varepsilon$ on the greedy action:

$$\begin{aligned} \sum_a \pi_\varepsilon(a \mid s) Q^\pi(s,a) &= \frac{\varepsilon}{|\mathcal{A}|} \sum_a Q^\pi(s,a) + (1-\varepsilon) \max_a Q^\pi(s,a) \\ &\ge \frac{\varepsilon}{|\mathcal{A}|} \sum_a Q^\pi(s,a) + (1-\varepsilon) \sum_a \pi(a \mid s) Q^\pi(s,a) \\ &\ge \sum_a \pi(a \mid s) Q^\pi(s,a) = V^\pi(s), \end{aligned} \tag{6}$$

where the first inequality uses $\max_a Q^\pi(s,a) \ge \sum_a \pi(a \mid s) Q^\pi(s,a)$ and the second uses $\varepsilon/|\mathcal{A}| \le \pi(a \mid s)$ for the non-greedy actions under a sufficiently exploratory policy. More precisely, the improvement follows from convexity: the ε-greedy policy shifts probability mass toward higher Q-values. □

Definition 4.3 (GLIE Condition)**.** A sequence of policies $\{\pi_t\}_{t=0}^{\infty}$ is *Greedy in the Limit with Infinite Exploration (GLIE)* if:

1. Every state–action pair is visited infinitely often: $\sum_{t=0}^{\infty} \mathbf{1}\{S_t = s, A_t = a\} = \infty$ almost surely for all $(s, a) \in \mathcal{S} \times \mathcal{A}$.
2. The policy converges to the greedy policy: $\pi_t(a \mid s) \to \mathbf{1}\{a = \arg\max_{a'} Q(s, a')\}$ almost surely for all $s \in \mathcal{S}$.

Example 4.4 (Standard GLIE Schedule)**.** The ε_t-greedy policy with $\varepsilon_t = 1/t$ satisfies GLIE. Condition 2 holds since $\varepsilon_t \to 0$. For Condition 1, since $\varepsilon_t > 0$ for all t and the MDP is finite, every state–action pair reachable under any positive-probability policy is visited infinitely often. Another common choice is $\varepsilon_t = c/(c + t)$ for some constant $c > 0$.

Constant ε Does Not Satisfy GLIE

Using a fixed $\varepsilon > 0$ violates Condition 2 of GLIE: the policy never becomes fully greedy. With constant ε, SARSA converges not to Q^* but to Q^{π_ε}, the value function of the ε-greedy policy. The gap $\|Q^* - Q^{\pi_\varepsilon}\|_\infty$ is bounded by $O(\varepsilon)$ but does not vanish.

With the exploration framework established, we now define the SARSA algorithm.

5 The SARSA Algorithm

Definition 5.1 (SARSA Update Rule)**.** Given the quintuple $(S_t, A_t, R_{t+1}, S_{t+1}, A_{t+1})$ where actions are drawn from the current ε-greedy policy $A_t \sim \pi_t(\cdot \mid S_t)$ and $A_{t+1} \sim \pi_t(\cdot \mid S_{t+1})$, the *SARSA update* is

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha_t\big[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\big]. \tag{7}$$

The name "SARSA" is an acronym for the quintuple $(\mathbf{S}_t, \mathbf{A}_t, \mathbf{R}_{t+1}, \mathbf{S}_{t+1}, \mathbf{A}_{t+1})$ used in the update.

Remark 5.2 (SARSA as TD(0) for Action Values). Comparing with the TD(0) update for state values (Lesson 47, Definition 4.2): $V(S_t) \leftarrow V(S_t) + \alpha[\delta_t]$ where $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$, the SARSA update is the direct analogue for action values with TD error

$$\delta_t = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t). \tag{8}$$

The only structural difference is that the bootstrap target uses the action-value at the *next state–action pair* rather than the next state alone.

SARSA: On-Policy TD Control

The SARSA algorithm performs GPI by interleaving single-step action-value evaluation (the TD update (7)) with policy improvement (selecting ε-greedy actions). It is *on-policy*: the action A_{t+1} in the bootstrap target is drawn from the *same* policy that the agent follows. This means SARSA learns the value of the policy it is actually executing, including the cost of exploratory actions.

Example 5.3 (SARSA Updates on a Grid)**.** Consider a 3×1 grid with states $\{L, C, R\}$, actions $\{\text{left}, \text{right}\}$, deterministic transitions (left/right move accordingly, with walls keeping the agent in place), reward $+1$ at R and 0 elsewhere, $\gamma = 0.9$, and $Q_0 \equiv 0$. Starting in C with $\varepsilon = 0.1$ and $\alpha = 0.1$:

Suppose the trajectory begins $S_0 = C, A_0 = \text{right}$ (greedy or exploratory), $R_1 = 1$, $S_1 = R, A_1 = \text{left}$ (action chosen by π_0). Then:

$$Q_1(C, \text{right}) = 0 + 0.1[1 + 0.9 \cdot Q_0(R, \text{left}) - 0] = 0.1[1 + 0] = 0.1.$$

After several episodes, $Q(C, \text{right})$ grows toward $Q^*(C, \text{right}) = 1 + 0.9 \cdot 1/(1 - 0.9) = 10$.

On-Policy Safety in Cliff Walking

The classic cliff-walking example [Sutton and Barto, 2018, Section 6.5] illustrates a key consequence of SARSA's on-policy nature. In a gridworld with a cliff along the bottom edge (falling off gives -100 reward), SARSA learns to take a *safe path* far from the cliff because its value estimates account for the ε-probability of exploratory actions that could step off the edge. Q-learning, being off-policy, learns the *optimal path* along the cliff edge — which is optimal only if the agent executes the greedy policy perfectly, without exploration. During learning, Q-learning's agent falls off the cliff much more frequently.

We now turn to the convergence analysis of SARSA.

6 Convergence of SARSA

Assumption 6.1 (SARSA Convergence Conditions).

1. The MDP has finite state space $\mathcal{S}$ and finite action space $\mathcal{A}$.
2. Rewards are bounded: $|r(s, a, s')| \leq R_{\max}$ for all (s, a, s').
3. The policy sequence $\{\pi_t\}$ satisfies GLIE (Definition 4.3).
4. Step sizes satisfy $\sum_n \alpha_n(s, a) = \infty$ and $\sum_n \alpha_n^2(s, a) < \infty$ for each $(s, a) \in \mathcal{S} \times \mathcal{A}$.

Convergence of SARSA (Singh et al. 2000)

Theorem 6.2 (Convergence of SARSA). *Under Assumption 6.1, the SARSA iterates converge to the optimal action-value function almost surely:*

$$Q_t(s, a) \xrightarrow{a.s.} Q^*(s, a) \quad \textit{for all } (s, a) \in \mathcal{S} \times \mathcal{A}. \tag{9}$$

Proof. We follow the approach of Singh et al. [2000], decomposing the SARSA update into a Q-learning component plus a vanishing exploration perturbation.

Step 1: Decompose the SARSA target. At time t, the SARSA update target is $R_{t+1} + \gamma Q_t(S_{t+1}, A_{t+1})$. Add and subtract $\gamma \max_{a'} Q_t(S_{t+1}, a')$:

$$R_{t+1} + \gamma Q_t(S_{t+1}, A_{t+1}) = \underbrace{R_{t+1} + \gamma \max_{a'} Q_t(S_{t+1}, a')}_{\text{Q-learning target}} + \underbrace{\gamma \big[Q_t(S_{t+1}, A_{t+1}) - \max_{a'} Q_t(S_{t+1}, a')\big]}_{\xi_t}. \tag{10}$$

The perturbation ξ_t captures the difference between following the current ε_t-greedy policy and acting greedily.

Step 2: Express as stochastic approximation with perturbation. At the k-th visit to state–action pair (s, a), the SARSA update can be written as

$$Q_{k+1}(s, a) = Q_k(s, a) + \alpha_k(s, a)\big[(\mathcal{T}_Q^* Q_k)(s, a) - Q_k(s, a) + w_k(s, a) + \xi_k(s, a)\big], \tag{11}$$

where $w_k(s, a)$ is the zero-mean stochastic approximation noise (the difference between the sampled Q-learning target and its expectation $(\mathcal{T}_Q^* Q_k)(s, a)$), and $\xi_k(s, a)$ is the exploration perturbation from Step 1.

Step 3: Verify that the noise is a martingale difference. Let $\mathcal{F}_k$ denote the filtration containing all information up to the k-th visit to (s, a). The noise $w_k(s, a) = R_{t+1} + \gamma \max_{a'} Q_k(S_{t+1}, a') - (\mathcal{T}_Q^* Q_k)(s, a)$ satisfies

$$\mathbb{E}[w_k(s, a) \mid \mathcal{F}_k] = (\mathcal{T}_Q^* Q_k)(s, a) - (\mathcal{T}_Q^* Q_k)(s, a) = 0, \tag{12}$$

by the definition of $\mathcal{T}_Q^*$. The variance is bounded by $\mathbb{E}[w_k^2 \mid \mathcal{F}_k] \le C$ for a constant C depending on $R_{\max}$ and γ, since rewards and Q-iterates are bounded (the latter by $R_{\max}/(1-\gamma)$, established by induction using the contraction property).

Step 4: Bound the exploration perturbation. Under the ε_t-greedy policy π_t, the conditional expectation of ξ_t given $\mathcal{F}_k$ and S_{t+1} is:

$$\begin{aligned}\mathbb{E}[\xi_t \mid \mathcal{F}_k, S_{t+1}] &= \gamma \sum_{a'} \pi_t(a' \mid S_{t+1}) Q_k(S_{t+1}, a') - \gamma \max_{a'} Q_k(S_{t+1}, a') \\ &= \gamma \left[(1-\varepsilon_t) \max_{a'} Q_k(S_{t+1}, a') + \frac{\varepsilon_t}{|\mathcal{A}|} \sum_{a'} Q_k(S_{t+1}, a') - \max_{a'} Q_k(S_{t+1}, a') \right] \\ &= \gamma\, \varepsilon_t \left[\frac{1}{|\mathcal{A}|} \sum_{a'} Q_k(S_{t+1}, a') - \max_{a'} Q_k(S_{t+1}, a') \right]. \end{aligned} \tag{13}$$

Since Q_k is bounded by $R_{\max}/(1-\gamma)$:

$$|\mathbb{E}[\xi_t \mid \mathcal{F}_k, S_{t+1}]| \le \gamma\, \varepsilon_t \cdot \frac{2R_{\max}}{1-\gamma} =: C_\xi\, \varepsilon_t. \tag{14}$$

Step 5: Apply perturbed stochastic approximation. The iteration (11) is a perturbed stochastic approximation of the Bellman optimality equation with driving function $h(Q)(s,a) = (\mathcal{T}_Q^* Q)(s,a) - Q(s,a)$. The unperturbed iteration (Q-learning) converges to Q^* by the Robbins–Monro theorem, since $\mathcal{T}_Q^*$ is a γ-contraction with unique fixed point Q^* (Lemma 3.5).

For the perturbation, we need $\sum_k \alpha_k(s,a)|\mathbb{E}[\xi_k \mid \mathcal{F}_k]| < \infty$ almost surely. By (14):

$$\sum_k \alpha_k(s,a)|\mathbb{E}[\xi_k \mid \mathcal{F}_k]| \le C_\xi \sum_k \alpha_k(s,a)\, \varepsilon_{t_k}, \tag{15}$$

where t_k is the time of the k-th visit to (s,a). Under the GLIE condition, $\varepsilon_t \to 0$. For the standard schedule $\alpha_n = 1/n$ and $\varepsilon_n = 1/n$: $\sum_k \alpha_k \varepsilon_{t_k} \le \sum_n 1/n^2 < \infty$. More generally, the GLIE condition requires ε_t to decrease fast enough that the perturbation series converges.

By the perturbed Robbins–Monro theorem: if the unperturbed stochastic approximation converges (which it does — this is Q-learning) and the perturbation satisfies $\sum_k \alpha_k |\mathbb{E}[\xi_k \mid \mathcal{F}_k]| < \infty$ a.s., then the perturbed iteration converges to the same limit.

Since $\mathcal{T}_Q^*$ is a γ-contraction with unique fixed point Q^*, and the perturbation is summable, the error $\|Q_t - Q^*\|_\infty$ converges to zero almost surely. □

Remark 6.3 (The Role of GLIE). The GLIE condition serves two purposes in the convergence proof. Condition 1 (infinite exploration) ensures all state–action pairs are visited infinitely often, which is needed for the step-size conditions to be meaningful. Condition 2 ($\varepsilon_t \to 0$) ensures that the exploration perturbation ξ_t vanishes, so that SARSA asymptotically behaves like Q-learning. Without Condition 2, the perturbation remains $O(\varepsilon)$ and SARSA converges to Q^{π_ε} rather than Q^*.

Example 6.4 (Verifying the Perturbation Bound)**.** Consider an MDP with $|\mathcal{A}| = 4$, $R_{\max} = 1$, $\gamma = 0.9$. Then $C_\xi = 0.9 \cdot 2/(1-0.9) = 18$. With $\varepsilon_{100} = 0.01$, the perturbation magnitude at step 100 is bounded by $18 \cdot 0.01 = 0.18$, a small fraction of the typical Q-value scale $R_{\max}/(1-\gamma) = 10$. As $\varepsilon_t \to 0$, this bound vanishes.

SARSA Convergence via Two-Timescale Analysis

An alternative view of SARSA convergence uses two-timescale stochastic approximation (Lesson 32). If the exploration rate ε_t decreases on a *faster* timescale than the step size α_t (i.e., $\varepsilon_t/\alpha_t \to 0$), then the policy converges to greedy "fast" while the Q-values update

"slowly," and the system tracks the equilibrium of Q-learning. This perspective, developed by Borkar [2008], provides a more general framework for analyzing policy improvement algorithms with diminishing exploration.

7 On-Policy Behavior of SARSA

Proposition 7.1 (SARSA with Fixed ε)**.** *If the exploration rate is held constant at $\varepsilon > 0$ (so the GLIE condition is violated), and all other conditions of Assumption 6.1 hold, then SARSA converges to Q^{π_ε}, the action-value function of the ε-greedy policy, rather than Q^*.*

Proof. With fixed ε, the SARSA update targets $\mathcal{T}_Q^{\pi_\varepsilon} Q$, the Bellman expectation operator of the ε-greedy policy. Since the policy is now fixed, SARSA is simply TD(0) applied to action values under π_ε. By the convergence of TD(0) (Lesson 47, Theorem 4.7, adapted to action values): $Q_t \to Q^{\pi_\varepsilon}$ almost surely. □

Example 7.2 (On-Policy vs. Off-Policy in a Stochastic Grid)**.** Consider a 4×4 grid with stochastic wind: with probability 0.3, the agent is pushed one cell upward regardless of its action. Near a cliff at the top row, the greedy-optimal path runs along the top (shortest distance to the goal), but exploratory actions during learning risk falling off. SARSA with $\varepsilon = 0.1$ learns to avoid the top row, assigning lower Q-values to states near the cliff because it accounts for the 10% chance of a random exploratory action pushing the agent over the edge. The converged $Q^{\pi_{0.1}}$ values reflect the true cost of the exploration policy, yielding a safer learned behavior.

This on-policy property distinguishes SARSA from Q-learning (Lesson 50) and is the principal reason for choosing SARSA in safety-critical applications.

8 Exercises

Exercise 8.1 (SARSA TD Error Properties)**.** (∗)
Let $\delta_t = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)$ be the SARSA TD error. If $Q = Q^\pi$ for the current (stationary) policy π, show that $\mathbb{E}_\pi[\delta_t \mid S_t, A_t] = 0$.

Exercise 8.2 (SARSA on a Three-State MDP)**.** (∗)
Consider $\mathcal{S} = \{A, B, C\}$ with $\mathcal{A} = \{1, 2\}$, deterministic transitions: action 1 in A leads to B, action 2 in A leads to C, all actions in B and C lead to A. Rewards: $r(A, 1) = 0$, $r(A, 2) = 1$, $r(B, \cdot) = 0$, $r(C, \cdot) = 2$. Let $\gamma = 0.5$, $\alpha = 0.5$, $Q_0 \equiv 0$, $\varepsilon = 0.1$. Suppose the first episode generates the trajectory $(A, 2, 1, C, 1, 0, A)$ and then $(A, 1, 0, B, 2, 0, A)$. Compute the SARSA updates for each step.

Exercise 8.3 (GLIE Schedule Design)**.** (∗∗)
Prove that the schedule $\varepsilon_t = c/(c + t)$ for $c > 0$ satisfies GLIE on any finite, communicating MDP. Specifically:

(a) Show that $\varepsilon_t \to 0$ (Condition 2).
(b) Show that for any (s, a), the probability that (s, a) is visited at time t is bounded below by a constant times $\varepsilon_t/|\mathcal{A}|$, and conclude from $\sum_t \varepsilon_t = \infty$ that (s, a) is visited infinitely often (Condition 1).

Exercise 8.4 (Perturbation Bound Tightness)**.** (∗∗)
In the convergence proof of Theorem 6.2, the perturbation bound (14) uses the worst case $|\mathcal{A}|^{-1} \sum_{a'} Q(s', a') - \max_{a'} Q(s', a')$.

(a) Show that this quantity is always non-positive and bounded in magnitude by $2R_{\max}/(1-\gamma)$.
(b) Give an example where the bound is tight (achieved with equality).

(c) Show that the bound can be improved to $R_{\max}(|\mathcal{A}|-1)/((1-\gamma)|\mathcal{A}|)$ by a more careful analysis.

Exercise 8.5 (SARSA with Fixed ε: Value Gap)**.** $(***)$
Let Q^* denote the optimal action-value function and Q^{π_ε} the action-value function of the ε-greedy policy with respect to Q^*. Show that

$$\|Q^* - Q^{\pi_\varepsilon}\|_\infty \le \frac{2\varepsilon\, R_{\max}}{(1-\gamma)^2}.$$

Hint: Use the simulation lemma: for any two policies π_1, π_2, the value difference satisfies $\|Q^{\pi_1} - Q^{\pi_2}\|_\infty \le \frac{1}{1-\gamma}\max_s \sum_a |\pi_1(a|s) - \pi_2(a|s)| \cdot R_{\max}/(1-\gamma)$.

Exercise 8.6 (SARSA Convergence Without GLIE)**.** $(***)$
Suppose $\varepsilon_t = \varepsilon > 0$ is constant, but we modify the SARSA update to use a decaying "effective exploration" $\tilde{\varepsilon}_t \to 0$ only in the bootstrap target while still executing the ε-greedy policy for data collection. Formally, the update becomes:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha_t\big[R_{t+1} + \gamma Q(S_{t+1}, \tilde{A}_{t+1}) - Q(S_t, A_t)\big],$$

where $\tilde{A}_{t+1}$ is drawn from a $\tilde{\varepsilon}_t$-greedy policy. Prove that this modified SARSA converges to Q^* and identify the sense in which it becomes off-policy.

9 Summary

- **From prediction to control**: GPI alternates between policy evaluation and improvement; model-free control requires learning action-value functions $Q(s,a)$ rather than state values $V(s)$.
- The **ε-greedy policy** assigns probability $1-\varepsilon+\varepsilon/|\mathcal{A}|$ to the greedy action and $\varepsilon/|\mathcal{A}|$ to all others, ensuring exploration while approximately following the greedy policy.
- The **GLIE condition** requires infinite exploration of all state–action pairs and convergence of the policy to greedy in the limit, typically achieved by $\varepsilon_t \to 0$.
- The **SARSA update** $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[\delta_t]$ with TD error $\delta_t = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)$ applies TD(0) to action values, using the *actually selected* next action A_{t+1}.
- **SARSA converges to** Q^* under GLIE (Theorem 6.2), proved by decomposing the update into a Q-learning component plus a vanishing exploration perturbation that is summable under $\varepsilon_t \to 0$.
- SARSA is **on-policy**: it evaluates the policy it actually follows, including exploration. With fixed ε, it converges to Q^{π_ε} rather than Q^*, which can yield safer behavior in environments where exploratory actions are costly.
- The action-value Bellman operators $\mathcal{T}_Q^\pi$ and $\mathcal{T}_Q^*$ are both γ-contractions in $\|\cdot\|_\infty$ (Lemma 3.5), providing the stability condition for all TD control convergence proofs.

10 Further Reading

- **Lesson 50** develops Q-learning (off-policy TD control), Expected SARSA, and provides a unified view of all three TD control algorithms. The comparison between SARSA and Q-learning deepens the on-policy vs. off-policy distinction introduced here.

- Singh et al. [2000] is the primary reference for the SARSA convergence proof presented in Theorem 6.2. Their framework extends to general single-step on-policy algorithms satisfying GLIE.
- Sutton and Barto [2018, Section 6.5] presents the cliff-walking example, the most widely cited illustration of the behavioral difference between on-policy SARSA and off-policy Q-learning.
- Rummery and Niranjan [1994] introduced SARSA (under the name "modified connectionist Q-learning") as an online alternative to Q-learning for use with neural network function approximators.
- **SARSA(λ)** combines SARSA with eligibility traces from Lesson 48, yielding faster credit assignment. See Sutton and Barto [2018, Chapter 12] for the algorithm and Bertsekas and Tsitsiklis [1996, Chapter 5] for convergence analysis.
- For the extension to continuous state spaces with function approximation, see Lessons 51–53. The "deadly triad" of function approximation, bootstrapping, and off-policy learning is less severe for on-policy SARSA than for off-policy Q-learning.

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